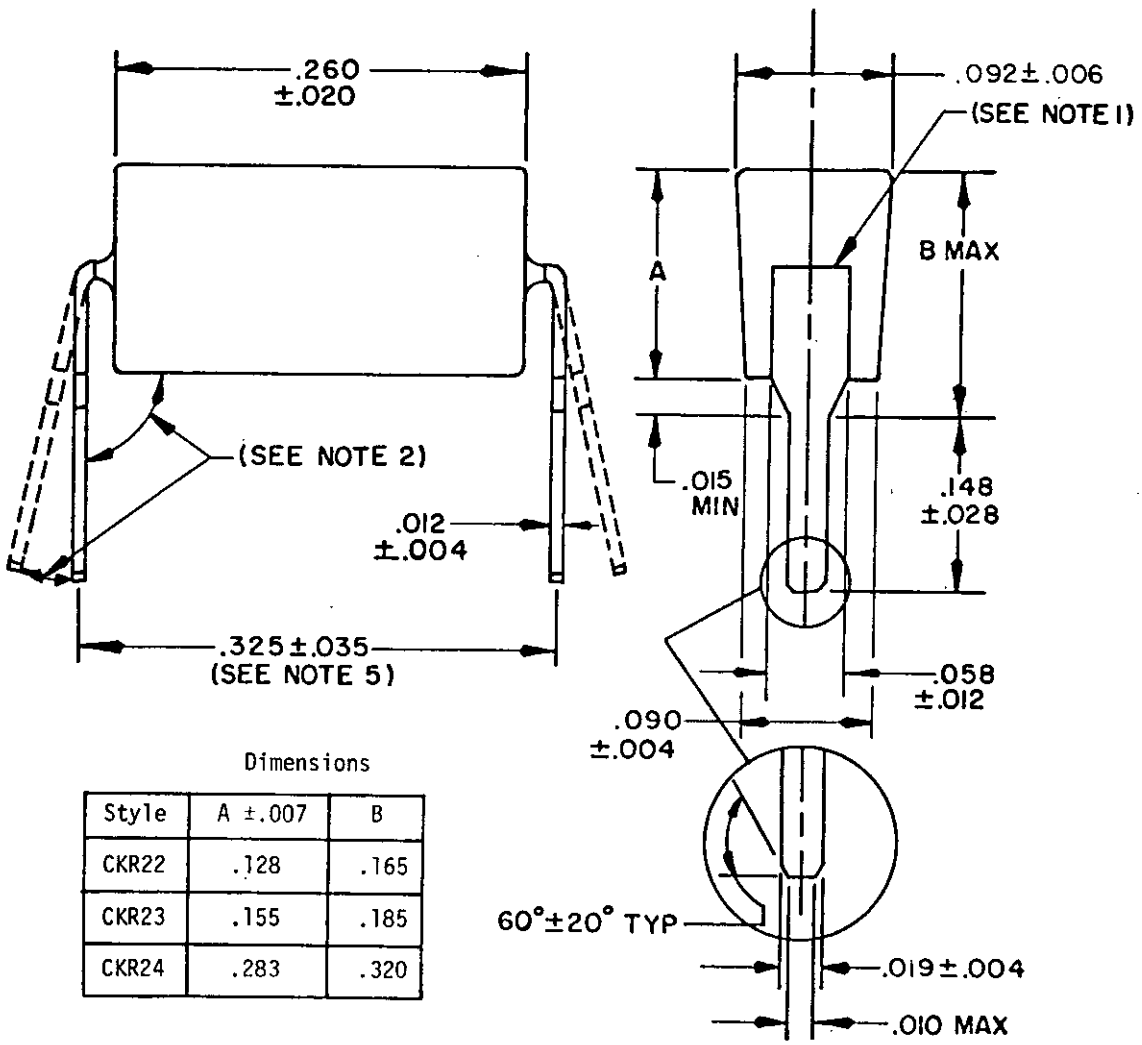
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), ESTABLISHED RELIABILITY, STYLES CKR22, CKR23, AND CKR24

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-39014.



Dimensions

Style	A ±.007	B
CKR22	.128	.165
CKR23	.155	.185
CKR24	.283	.320

FIGURE 1. Style CKR22, CKR23, and CKR24 Capacitors.

Ⓑ denotes changes

FSC 5910

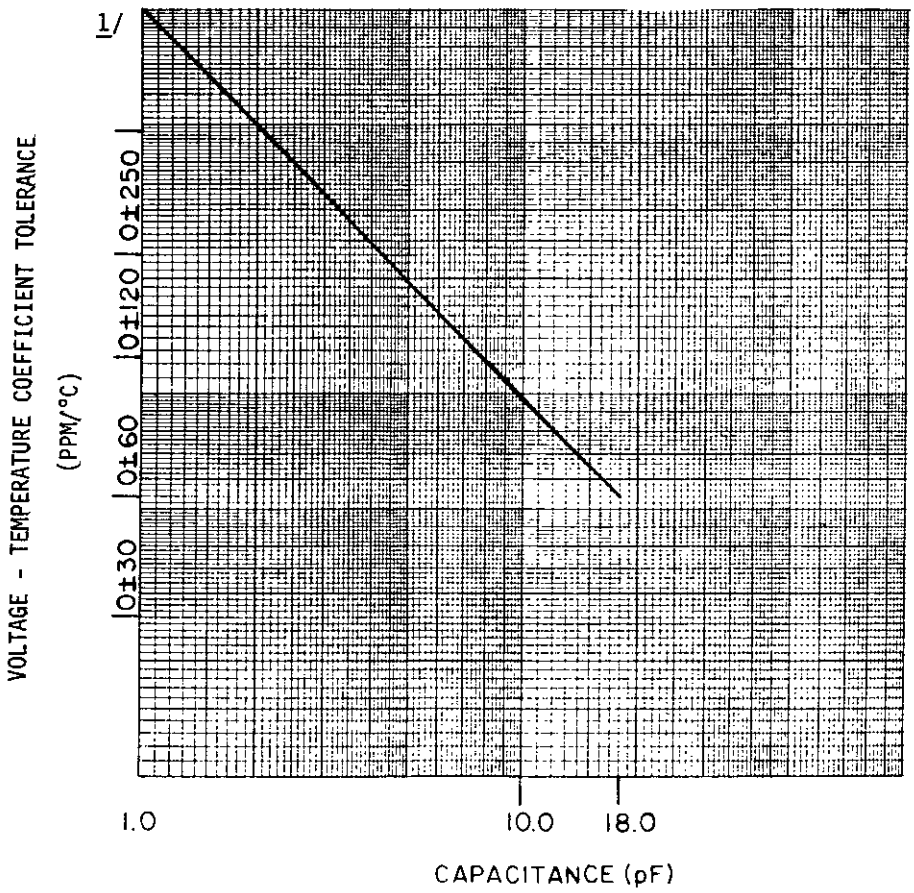
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INCHES	MM	INCHES	MM
.004	.10	.090	2.29
.006	.15	.092	2.34
.007	.18	.128	3.25
.010	.25	.148	3.76
.012	.30	.155	3.94
.015	.38	.165	4.19
.019	.48	.185	4.70
.020	.51	.260	6.60
.028	.71	.283	7.19
.035	.89	.320	8.13
.058	1.47	.325	8.26

NOTES:
1. Leads shall be centered within ±.005 inches (.13 mm).
2. The angle shall be 95° +10° -5°.
3. Dimensions are in inches.
4. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
5. The distance between the centers of the mounting holes will be .300 ±.010 inches (7.62 ±.25 mm).
6. Nonconductive material shall not extend beyond .030 inches (.76 mm) from the edge of the capacitor body.

FIGURE 1. Style CKR22, CKR23, and CKR24 capacitors - Continued.

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1/ Characteristics of capacitors with values of 2.1 pF or less are not considered practicably measureable.

Ⓑ

FIGURE 2. Voltage-temperature limits for capacitance values of 18 pF or less.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Molded plastic encapsulated.

Failure rate level: M, P, R, and S.

Capacitance value: See table II.

Capacitance tolerance: D = ±.5 pF, F = ±1%, J = ±5%, K = ±10%, and M = ±20%.

Operating temperature range: -55° to +125°C.

Voltage rating: See table II.

Dissipation factor: See table I.

Ⓑ Dielectric withstanding voltage: Body insulation shall be able to withstand 500 Vdc between leads and case.

Insulation resistance:

At 25°C - Not less than 100,000 megohms or 1,000 megohm-microfarads, whichever is less.

At 125°C - Not less than 10,000 megohms or 100 megohm-microfarads, whichever is less.

Insertion loss: Not applicable.

Voltage-temperature limits: See table I. For capacitors with values of 18 pF or less, the voltage-temperature limits tolerance shall be determined in accordance with the curve shown on figure 2.

Ⓑ TABLE I. Dissipation factors and voltage-temperature limits.

Capacitance change with reference to 25°C			
Characteristic symbol	Steps A through D table V of MIL-C-39014 Bias = 0 volts	Steps E through G table V of MIL-C-39014 Bias = rated voltage	Dissipation factor (percent)
CG CH	0 ±30 ppm/°C 1/ 0 ±60 ppm/°C 1/	0 ±30 ppm/°C 1/ 0 ±60 ppm/°C 1/	.15 .15
BX	±15 percent	+15, -25 percent as applicable	2.5
BR	±15 percent	+15, -40 percent as applicable	3.0

1/ See figure 2 for capacitance value less than 20 pF.

Immersion:
IR - Not less than 50 percent of initial 25°C requirements.

Resistance to soldering heat:

Ⓑ ΔCap: ±5 percent or .25 pF, whichever is greater for CG and CH characteristics.
+20, -0 percent for characteristics BX and BR.

DF - Not to exceed the initial limit.

IR - Not less than initial 25°C limit.

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Terminal strength: MIL-STD-202, method 211, condition B, except that each lead shall be bent away from the body 90° from the original position (see figure 1) and back, two bends.

Life:

Rated conditions, 100 percent of dc rated voltage at 125°C.

0 through 3,000 hours:

IR - At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.

Ⓑ ΔC - ±5 percent or .5 pF, whichever is greater from initial measured value for CG and CH characteristics.
±20 percent from initial measured value for characteristics BX and BR.

Dissipation factor - Not to exceed the initial limit.

4,000 through 32,000 hours:

IR - At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.

Ⓑ ΔC - ±5 percent or .5 pF, whichever is greater from initial measured value for CG and CH characteristics.
±20 percent from initial measured value for characteristics BX and BR.

Dissipation factor - Not to exceed the initial limit.

Accelerated condition, 200 percent of dc rated voltage at 125°C.

0 and 250 hours:

IR - At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.

Ⓑ ΔC - ±5 percent or .5 pF, whichever is greater from initial measured value for CG and CH characteristics.
±20 percent from initial measured value for characteristics BX and BR.

Dissipation factor - Not to exceed the initial limit.

Ⓑ 1,000, 2,000, and 4,000 hours:

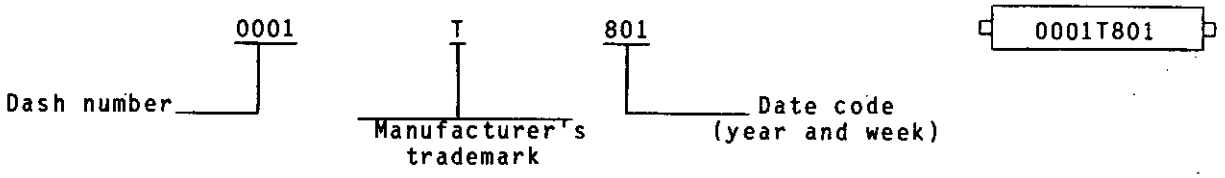
IR - At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.

ΔC - ±20 percent from initial measured value.

Dissipation factor - Not to exceed the initial limit.

Part number: M39014/22- (dash numbers from table II), i.e. M39014/22-0001.

Marking: In accordance with MIL-C-39014, except that minimum marking as shown in the following example, shall be used. Full marking shall be included on the package.



Marking example

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TABLE II. Dash numbers and electrical characteristics.

Dash number Failure rate level (%/1,000 hours)				Capacitance (pF)	Capacitance tolerance	DC rated voltage (volts)
I.0(M)	0.1(P)	0.01(R)	0.001(S)			
Style CKR22, see figure 2 for voltage-temperature limits.						
0001	0301	0601	0901	1.0	D	200
0004	0304	0604	0904	1.2	D	
0007	0307	0607	0907	1.5	D	
0010	0310	0610	0910	1.8	D	
0013	0313	0613	0913	2.2	D	
0016	0316	0616	0916	2.7	D	
0019	0319	0619	0919	3.3	D	
0022	0322	0622	0922	3.9	D	
0025	0325	0625	0925	4.7	D	
0028	0328	0628	0928	5.6	D	
0031	0331	0631	0931	6.8	D	
0034	0334	0634	0934	8.2	D	
0037	0337	0637	0937	10	D	
0038	0338	0638	0938	10	J	
0039	0339	0639	0939	10	K	
0040	0340	0640	0940	12	D	
0041	0341	0641	0941	12	J	
0042	0342	0642	0942	12	K	
0043	0343	0643	0943	15	D	
0044	0344	0644	0944	15	J	
0045	0345	0645	0945	15	K	
0046	0346	0646	0946	18	D	
0047	0347	0647	0947	18	J	
0048	0348	0648	0948	18	K	
Style CKR22, Voltage-temperature limits of 0 ±30 ppm/°C.						
0049	0349	0649	0949	22	D	200
0050	0350	0650	0950	22	J	
0051	0351	0651	0951	22	K	
0052	0352	0652	0952	27	D	
0053	0353	0653	0953	27	J	
0054	0354	0654	0954	27	K	
0055	0355	0655	0955	33	D	
0056	0356	0656	0956	33	J	
0057	0357	0657	0957	33	K	
0058	0358	0658	0958	39	D	
0059	0359	0659	0959	39	J	
0060	0360	0660	0960	39	K	
0061	0361	0661	0961	47	D	
0062	0362	0662	0962	47	J	
0063	0363	0663	0963	47	K	
0064	0364	0664	0964	56	D	
0065	0365	0665	0965	56	J	
0066	0366	0666	0966	56	K	
0067	0367	0667	0967	68	F	
0068	0368	0668	0968	68	J	
0069	0369	0669	0969	68	K	
0070	0370	0670	0970	82	F	
0071	0371	0671	0971	82	J	
0072	0372	0672	0972	82	K	
0073	0373	0673	0973	100	F	
0074	0374	0674	0974	100	J	
0075	0375	0675	0975	100	K	
0076	0376	0676	0976	120	F	
0077	0377	0677	0977	120	J	
0078	0378	0678	0978	120	K	
0079	0379	0679	0979	150	F	

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TABLE II. Dash numbers and electrical characteristics - Continued.

Dash number Failure rate level (%/1,000 hours)				Capacitance (pF)	Capacitance tolerance	DC rated voltage (volts)
1.0(M)	0.1(P)	0.01(R)	0.001(S)			
Style CKR22, Voltage-temperature limits of 0 ±30 ppm/°C - Continued.						
0080	0380	0680	0980	150	J	200
0081	0381	0681	0981	150	K	
0082	0382	0682	0982	180	F	
0083	0383	0683	0983	180	J	
0084	0384	0684	0984	180	K	
0085	0385	0685	0985	220	F	
0086	0386	0686	0986	220	J	
0087	0387	0687	0987	220	K	
0088	0388	0688	0988	270	F	
0089	0389	0689	0989	270	J	
0090	0390	0690	0990	270	K	
0091	0391	0691	0991	330	F	
0092	0392	0692	0992	330	J	
0093	0393	0693	0993	330	K	
0094	0394	0694	0994	390	F	
0095	0395	0695	0995	390	J	
0096	0396	0696	0996	390	K	
0097	0397	0697	0997	470	F	
0098	0398	0698	0998	470	J	
0099	0399	0699	0999	470	K	
0100	0400	0700	1000	560	F	100
0101	0401	0701	1001	560	J	
0102	0402	0702	1002	560	K	
0103	0403	0703	1003	680	F	
0104	0404	0704	1004	680	J	
0105	0405	0705	1005	680	K	
0106	0406	0706	1006	820	F	
0107	0407	0707	1007	820	J	
0108	0408	0708	1008	820	K	
0109	0409	0709	1009	1,000	F	
0110	0410	0710	1010	1,000	J	
0111	0411	0711	1011	1,000	K	
0112	0412	0712	1012	1,200	F	
0113	0413	0713	1013	1,200	J	
0114	0414	0714	1014	1,200	K	
0115	0415	0715	1015	1,500	F	
0116	0416	0716	1016	1,500	J	
0117	0417	0717	1017	1,500	K	
0118	0418	0718	1018	1,800	F	
0119	0419	0719	1019	1,800	J	
0120	0420	0720	1020	1,800	K	
0121	0421	0721	1021	2,200	F	
0122	0422	0722	1022	2,200	J	
0123	0423	0723	1023	2,200	K	
0124	0424	0724	1024	2,700	F	50
0125	0425	0725	1025	2,700	J	
0126	0426	0726	1026	2,700	K	
0127	0427	0727	1027	3,300	F	
0128	0428	0728	1028	3,300	J	
0129	0429	0729	1029	3,300	K	
0130	0430	0730	1030	3,900	F	
0131	0431	0731	1031	3,900	J	
0132	0432	0732	1032	3,900	K	
0133	0433	0733	1033	4,700	F	
0134	0434	0734	1034	4,700	J	
0135	0435	0735	1035	4,700	K	
0136	0436	0736	1036	5,600	F	

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TABLE II. Dash numbers and electrical characteristics - Continued.

Dash number Failure rate level (%/1,000 hours)				Capacitance (pF)	Capacitance tolerance	DC rated voltage (volts)
1.0(M)	0.1(P)	0.01(R)	0.001(S)			
Style CKR22, Voltage-temperature limits of 0 ±30 ppm/°C - Continued.						
0137	0437	0737	1037	5,600	J	50
0138	0438	0738	1038	5,600	K	
0139	0439	0739	1039	6,800	F	
0140	0440	0740	1040	6,800	J	
0141	0441	0741	1041	6,800	K	
0142	0442	0742	1042	8,200	F	
0143	0443	0743	1043	8,200	J	
0144	0444	0744	1044	8,200	K	
0145	0445	0745	1045	10,000	F	
0146	0446	0746	1046	10,000	J	
0147	0447	0747	1047	10,000	K	
Style CKR22, Voltage-temperature limits of ±15% (and +15%, -25%)						
0148	0448	0748	1048	270	K	200
0149	0449	0749	1049	330	K	
0150	0450	0750	1050	330	M	
0151	0451	0751	1051	390	K	
0152	0452	0752	1052	470	K	
0153	0453	0753	1053	470	M	
0154	0454	0754	1054	560	K	
0155	0455	0755	1055	680	K	
0156	0456	0756	1056	680	M	
0157	0457	0757	1057	820	K	
0158	0458	0758	1058	1,000	K	100
0159	0459	0759	1059	1,000	M	
0160	0460	0760	1060	1,200	K	
0161	0461	0761	1061	1,500	K	
0162	0462	0762	1062	1,500	M	
0163	0463	0763	1063	1,800	K	
0164	0464	0764	1064	2,200	K	
0165	0465	0765	1065	2,200	M	
0166	0466	0766	1066	2,700	K	
0167	0467	0767	1067	3,300	K	
0168	0468	0768	1068	3,300	M	
0169	0469	0769	1069	3,900	K	
0170	0470	0770	1070	4,700	K	
0171	0471	0771	1071	4,700	M	
0172	0472	0772	1072	5,600	K	
0173	0473	0773	1073	6,800	K	
0174	0474	0774	1074	6,800	M	
0175	0475	0775	1075	8,200	K	
0176	0476	0776	1076	10,000	K	
0177	0477	0777	1077	10,000	M	
0178	0478	0778	1078	12,000	K	50
0179	0479	0779	1079	15,000	K	
0180	0480	0780	1080	15,000	M	
0181	0481	0781	1081	18,000	K	
0182	0482	0782	1082	22,000	K	
0183	0483	0783	1083	22,000	M	
0184	0484	0784	1084	27,000	K	
0185	0485	0785	1085	33,000	K	
0186	0486	0786	1086	33,000	M	
0187	0487	0787	1087	39,000	K	
0188	0488	0788	1088	47,000	K	
0189	0489	0789	1089	47,000	M	
0190	0490	0790	1090	56,000	K	
0191	0491	0791	1091	68,000	K	

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TABLE II. Dash numbers and electrical characteristics - Continued.

Dash number Failure rate level (%/1,000 hours)				Capacitance (pF)	Capacitance tolerance	DC rated voltage (volts)
1.0(M)	0.1(P)	0.01(R)	0.001(S)			
Style CKR22, Voltage-temperature limits of ±15% and +15%, -25% - Continued.						
0192	0492	0792	1092	68,000	M	50
0193	0493	0793	1093	82,000	K	
0194	0494	0794	1094	100,000	K	
0195	0495	0795	1095	100,000	M	
Style CKR23, Voltage-temperature limits of 0 ±30 ppm/°C						
0258	0558	0858	1158	560	F	200
0259	0559	0859	1159	560	J	
0260	0560	0860	1160	560	K	
0261	0561	0861	1161	680	F	
0262	0562	0862	1162	680	J	
0263	0563	0863	1163	680	K	
0264	0564	0864	1164	820	F	
0265	0565	0865	1165	820	J	
0266	0566	0866	1166	820	K	
0267	0567	0867	1167	1,000	F	
0268	0568	0868	1168	1,000	J	
0269	0569	0869	1169	1,000	K	
0270	0570	0870	1170	1,200	F	
0271	0571	0871	1171	1,200	J	
0272	0572	0872	1172	1,200	K	
0273	0573	0873	1173	2,700	F	100
0274	0574	0874	1174	2,700	J	
0275	0575	0875	1175	2,700	K	
0276	0576	0876	1176	3,300	F	
0277	0577	0877	1177	3,300	J	
0278	0578	0878	1178	3,300	K	
0279	0579	0879	1179	4,700	F	50
0280	0580	0880	1180	4,700	J	
0281	0581	0881	1181	4,700	K	
0282	0582	0882	1182	5,600	F	
0283	0583	0883	1183	5,600	J	
0284	0584	0884	1184	5,600	K	
0285	0585	0885	1185	6,800	F	
0286	0586	0886	1186	6,800	J	
0287	0587	0887	1187	6,800	K	
0288	0588	0888	1188	8,200	F	
0289	0589	0889	1189	8,200	J	
0290	0590	0890	1190	8,200	K	
0291	0591	0891	1191	10,000	F	
0292	0592	0892	1192	10,000	J	
0293	0593	0893	1193	10,000	K	
Style CKR23, Voltage-temperature limits of +15% and +15, -25% - Continued.						
0196	0496	0796	1096	1,000	K	200
0197	0497	0797	1097	1,000	M	
0198	0498	0798	1098	1,200	K	
0199	0499	0799	1099	1,500	K	
0200	0500	0800	1100	1,500	M	
0201	0501	0801	1101	1,800	K	
0202	0502	0802	1102	2,200	K	
0203	0503	0803	1103	2,200	M	
0204	0504	0804	1104	2,700	K	
0205	0505	0805	1105	3,300	K	
0206	0506	0806	1106	3,300	M	

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TABLE II. Dash numbers and electrical characteristics - Continued.

Dash number Failure rate level (%/1,000 hours)				Capacitance (pF)	Capacitance tolerance	DC rated voltage (volts)
1.0(M)	0.1(P)	0.01(R)	0.001(S)			
Style CKR23, Voltage-temperature limits of +15% and +15, -25%- Continued.						
0207	0507	0807	1107	3,900	K	200
0208	0508	0808	1108	4,700	K	
0209	0509	0809	1109	4,700	M	
0210	0510	0810	1110	5,600	K	
0211	0511	0811	1111	6,800	K	
0212	0512	0812	1112	6,800	M	
0213	0513	0813	1113	8,200	K	
0214	0514	0814	1114	10,000	K	
0215	0515	0815	1115	10,000	M	
0216	0516	0816	1116	12,000	K	100
0217	0517	0817	1117	15,000	K	
0218	0518	0818	1118	15,000	M	
0219	0519	0819	1119	18,000	K	
0220	0520	0820	1120	22,000	K	
0221	0521	0821	1121	22,000	M	
0222	0522	0822	1122	27,000	K	
0223	0523	0823	1123	33,000	K	
0224	0524	0824	1124	33,000	M	
0225	0525	0825	1125	39,000	K	
0226	0526	0826	1126	47,000	K	
0227	0527	0827	1127	47,000	M	
0228	0528	0828	1128	56,000	K	
0229	0529	0829	1129	68,000	K	
0230	0530	0830	1130	68,000	M	
0231	0531	0831	1131	82,000	K	
0232	0532	0832	1132	100,000	K	
0233	0533	0833	1133	100,000	M	
0234	0534	0834	1134	120,000	K	50
0235	0535	0835	1135	150,000	K	
0236	0536	0836	1136	150,000	M	
0237	0537	0837	1137	180,000	K	
0238	0538	0838	1138	220,000	K	
0239	0539	0839	1139	220,000	M	
Style CKR24, Voltage-temperature limits of ±15% and +15, -40%.						
0240	0540	0840	1140	120,000	K	100
0241	0541	0841	1141	150,000	K	100
0242	0542	0842	1142	150,000	M	100
0243	0543	0843	1143	180,000	K	50
0244	0544	0844	1144	220,000	K	
0245	0545	0845	1145	220,000	M	
0246	0546	0846	1146	270,000	K	
0247	0547	0847	1147	330,000	K	
0248	0548	0848	1148	330,000	M	
0249	0549	0849	1149	390,000	K	
0250	0550	0850	1150	470,000	K	
0251	0551	0851	1151	470,000	M	
0252	0552	0852	1152	560,000	K	
0253	0553	0853	1153	680,000	K	
0254	0554	0854	1154	680,000	M	
0255	0555	0855	1155	820,000	K	
0256	0556	0856	1156	1,000,000	K	
0257	0557	0857	1157	1,000,000	M	

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Custodians:
Army - ER
Navy - EC
Air Force - 11
NASA - NA

Review activities:
Army - MI, AR
Navy -
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5910-1416)

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C. REASON FOR RECOMMENDED CHANGE(S)

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